TIM 172B, Lecture # 9, MOT-II (SCM); 2/3/26

Agenda

1. HW # 4 : brief discussion
   work on HW # 4 ASAP!

2. Inventory Management : Formulas & Calculations

3. Inventory Management : Derivation of the formulas

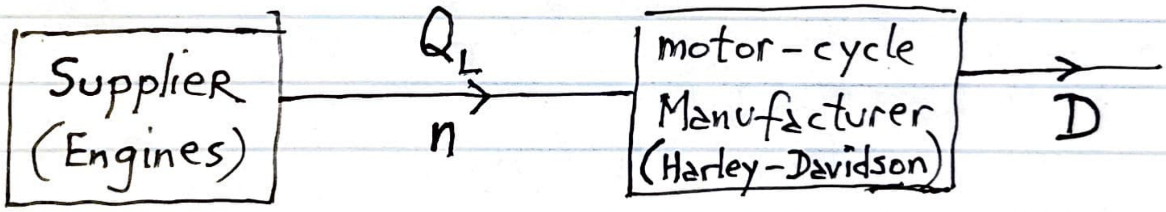
2. Inventory Management

Cycle Inventory :

Assumption : Safety Inventory = 0

Clearly define the context (using the stage representation and/or the SC Network representation.)

Within the above context (stage and/or Network), let look at the following sub-context (aka "the set-up")



Annual Demand $\triangleq D \dfrac{\text{units}}{\text{Year}}$ [from Demand forecasting]

$\left.\begin{array}{l}\text{\# of shipments or order} \\ \text{per YR. to fulfill annual demand}\end{array}\right\} \triangleq n \dfrac{\text{shipments}}{\text{Year}}$

\# of units per order or shipment $\triangleq Q_L \dfrac{\text{units}}{\text{shipment}}$

Q_L is called the lot size

$$\Rightarrow \quad (n)(Q_L) = D \longrightarrow (1)$$

$$\left[\frac{\text{shipments}}{\text{year}}\right]\left[\frac{\text{units}}{\text{shipment}}\right] = \left[\frac{\text{units}}{\text{year}}\right]$$

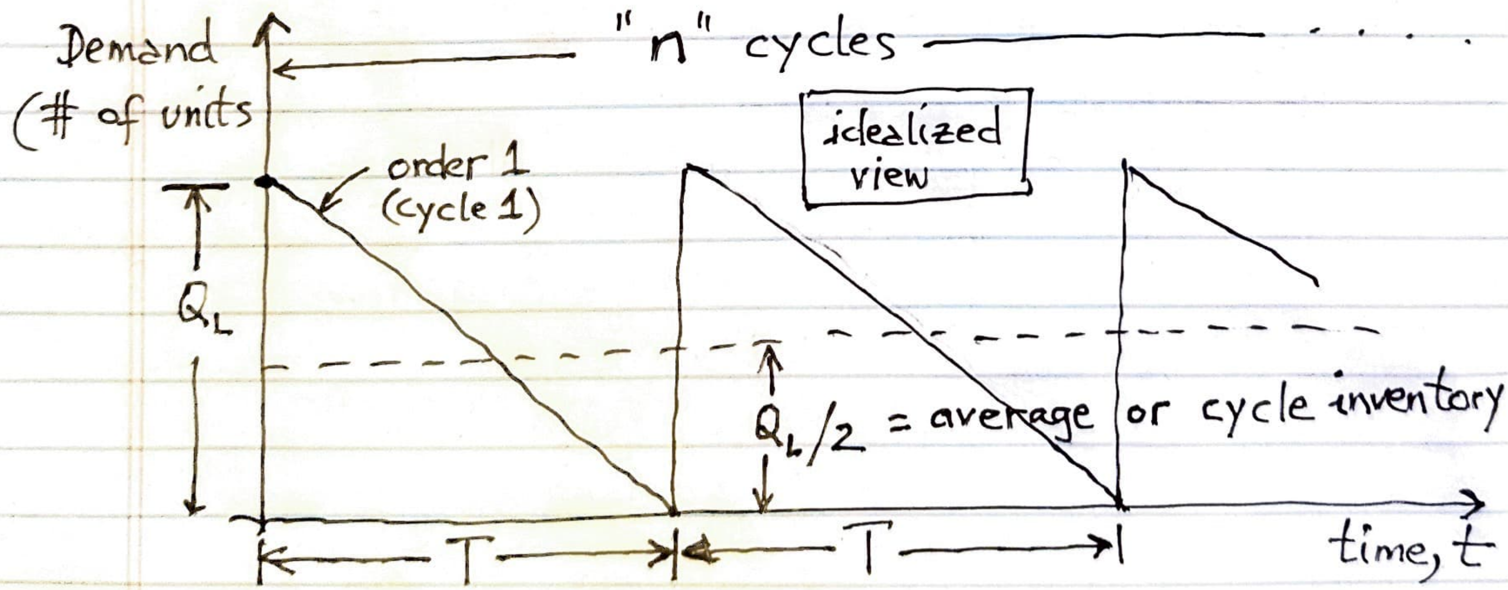
T is the time between 2 successive orders (or shipments), i.e., every T days the inventory needs to be <u>replenished</u>.

$T \triangleq$ inventory replenishment time, aka replenishment cycle time (days)

If there are "n" orders per year

$$(n)(T) = 365 \longrightarrow (2)$$

$$\left[\frac{\text{orders}}{\text{year}}\right]\left[\frac{\text{days}}{\text{order}}\right] = \left[\frac{\text{days}}{\text{year}}\right]$$



for The above diagram:

$$Q_L \triangleq \text{lot size} : \frac{\#\text{ of units}}{\text{order}}$$

$$T \triangleq \text{replenishment cycle time (days)}$$

Average inventory or Cycle Inventory $= \frac{Q_L}{2} \rightarrow (3)$

The Cycle Inventory Management problem :

Given : Annual demand, D,

determine the optimal lot size, Q_L^*, in order to minimize the total cost, C_T, (i.e. maximize efficiency).

Result :

The optimal lot size that minimizes total cost, C_T, is

$$Q_L^* = \sqrt{\frac{2DS}{hC}} , \longrightarrow (4)$$

where

$D \triangleq$ annual demand

$S =$ shipment cost per order

$h \triangleq$ % holding cost of holding \$1 of inventory for 1 year

Example : $h = 20\% \Rightarrow$ cost of holding \$1 (= 100 cents) for 1 year is \$0.20 (= 20 cents)

$C \triangleq$ cost of 1 unit (in our case, 1 engine)

$$\left.\begin{array}{l}\text{Cost of holding} \\ \text{1 unit of supply}\end{array}\right\} = hC \longrightarrow (5)$$

Back to our example

Engine supplier (Milwaukee) $\xrightarrow{\text{Trucks}}$ H-D Assembly plant, PA $\longrightarrow$ $D_d = 300 \frac{\text{cycles}}{\text{day}}$

Q_L, S

Engine cost $C = \$500$. per unit (engine)

$S = \$1000$. per trip or order

D_d : daily demand

$h = 20\% \equiv \$0.20$ per "\$ of inventory" per year

1. What is the optimal lot size, Q_L^*, for H-D ordering engines from its supplier

Assumption [for "in-class" only] : "slow economy

$12 \frac{\text{months}}{\text{year}} \times 4 \frac{\text{weeks}}{\text{month}} \times 5 \frac{\text{days}}{\text{week}}$

\# of working days per year $= 240 \frac{\text{days}}{\text{year}}$

$\Rightarrow$ Annual demand, $D = (D_d) \times 240$

$= 300 \left[\frac{\text{units}}{\text{day}}\right] \times 240 \left[\frac{\text{days}}{\text{year}}\right]$

$$\Rightarrow \quad D = 72{,}000 \ \frac{\text{units}}{\text{year}}$$

Optimal lot size, $Q_L^* \underset{(4)}{=} \sqrt{\dfrac{2DS}{hC}}$

$$= \sqrt{\frac{(2)(72{,}000)(1000)}{(0.20)(500)}}$$

$$= \sqrt{(2)(72{,}000)(10)}$$

$$= \sqrt{1440000}$$

(i) $Q_L^* = 1200 \ \dfrac{\text{units}}{\text{order}}$

Conclusion : For efficiency, or "economy of scale", or "minimizing total cost", H-D needs to order 1200 engines on each shipment or order.

(ii) Average or cycle inventory $\underset{(3)}{=} \dfrac{Q_L^*}{2} = 600$ units

$\Rightarrow$ H-D carries, on average (aka during 1 cycle) an inventory of 600 units (or engines) per year.

(iii) Order or shipment frequency $\left.\right\}$, $n^* \underset{(1)}{=} \dfrac{D}{Q_L^*} = \dfrac{72{,}000}{1200}$

$= 60 \dfrac{\text{orders}}{\text{year}}$

$\Rightarrow$ To satisfy or fulfill annual demand, H-D needs $60 \dfrac{\text{orders}}{\text{year}}$

or 60 shipments/year

(iv) Replenishment cycle time, $T \underset{(2)}{=} \dfrac{365}{n} = \dfrac{365}{60}$

≈ 6 days.

$\Rightarrow$ Inventory needs to be replenished every 6 days.

(v) Flow time : average amount of time that 1 unit of supply (i.e. 1 engine) is held in inventory

$= \dfrac{T}{2} \longrightarrow (6)$

$=$ 3 days in this case

$\Rightarrow$ Each engine, on average, is held in inventory, for 3 days.

(vi) $\left.\begin{array}{l}\text{The cycle inventory}\\ \text{holding cost } (\underline{\text{annual}})\end{array}\right\} = \left(\begin{array}{c}\text{Cycle}\\ \text{Inventory}\end{array}\right) \times \left(\begin{array}{c}\text{Cost of}\\ \text{holding 1 unit}\\ \text{for 1 year}\end{array}\right)$

$$C_I \underset{(3,5)}{=} \left(\frac{Q_L}{2}\right) \cdot (hC) \longrightarrow (7)$$

(vii) The minimum cycle inventory holding cost, $C_I^* = C_I \Big|_{Q = Q_L^*}$

$$C_I^* \underset{(7)}{=} \left(\frac{Q_L^*}{2}\right)(hC), \longrightarrow (8)$$

where Q_L^* is given by eqn. (4)

(viii) at the optimal lot size, Q_L^*, $C_S^* = C_I^*$,

where C_S^* is the minimum shipment or order cost.

3. Derivation of eqn (4) for the optimal lot size, Q_L^*

Objective : Determine the optimal lot size that minimizes (i.e., optimizes) the total cost, C_T (annual)

Process & Implementation

1. Determine the annual total cost, C_T

$$C_T = \underbrace{C_M}_{\text{material cost}} + \underbrace{C_S}_{\text{transportation cost}} + \underbrace{C_I}_{\text{cycle inventory holding cost}}$$

$$C_T \underset{(3,5)}{=} DC + nS + \underbrace{\left(\frac{Q_L}{2}\right)}_{\text{from eqn.(3)}} \underbrace{hC}_{\text{eqn (5)}}$$

$$C_T = DC + \underbrace{\left(\frac{D}{Q_L}\right)}_{\text{eqn (1)}} S + \left(\frac{Q_L}{2}\right) hC = f(Q_L) \rightarrow (9)$$

2. To determine the lot size, Q_L, that minimizes total cost,

$$\frac{dC_T}{dQ_L} = 0 \Rightarrow Q_L^*$$, the lot size that minimizes C_T